Arun Satish Rao

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PROFILE

Computer science educator and Director of Robotics at Stratford Preparatory Blackford, pursuing doctoral research in computer science education. Experience spans AP computer science and cybersecurity teaching, adapting university curricula for high school learners, and leading competitive robotics programs in a program-management role.

RESEARCH INTERESTS

K-12 computer science education; AI literacy and AI-supported learning in schools; cybersecurity education; robotics education and competition-based learning; design and evaluation of learning technologies.

EDUCATION

M.S., Computer Science May 2024
California State University, East Bay, Hayward, CA - GPA: 3.636
B.S., Computer Science and Engineering May 2022
NMAM Institute of Technology (NMAMIT), Karnataka, India

TEACHING AND LEADERSHIP EXPERIENCE

Stratford Preparatory Blackford, San Jose, CA
Director of Robotics Aug 2026 - Present
Computer Science Teacher Jul 2025 - Present

- Direct a four-team robotics program (the flagship varsity FRC team and three FTC teams: one varsity, two junior varsity) in a program and project management role: coordinate the mechanical, electronics, software, design, and business sub-teams, lead outreach, and guide the technical teams and their mentors.
- Teach AP Computer Science A (2025-26 and 2026-27), AP Cybersecurity, and Honors Robotics.
- Designed and taught an Advanced Cybersecurity course adapted from UC Berkeley's CS 161 (Computer Security) undergraduate syllabus, re-scoped and re-sequenced for high school rigor and student level (2025-26).
- Coached the varsity and junior varsity FTC teams through league play and qualifiers (2025-26); the teams reached the playoffs in their first competition season.
- Advisor, Cybersecurity Club: mentor students preparing for and competing in capture-the-flag competitions, including PicoCTF.

Oakland Unified School District, Oakland, CA
Computer Science Instructor Aug 2024 - Jun 2025

- Designed and delivered a curriculum covering programming, data structures, and algorithms, with hands-on projects and interactive JavaScript and Code.org lessons.
- Mentored senior students in capstone projects on web development and research methodologies, resulting in student research papers and technical portfolios.

California State University, East Bay, Hayward, CA
Teaching Associate and Instructional Student Assistant Aug 2023 - May 2024

- Led hands-on C++ lab sessions for CS 201 (assignment scores improved 25%); tutored mathematics and computer science one-on-one at the Student Center for Academic Achievement and helped build self-paced digital learning resources.

INDUSTRY EXPERIENCE

One Community Global, Remote
Volunteer Software Engineer Jun 2024 - Oct 2024

- Pull request reviews and code quality work contributed to a 15% reduction in production-breaking bugs; performance testing and optimizations improved load times by 20%.

BlueLine Computers, Karnataka, India
Android App Developer Intern Jun 2021 - Aug 2021

- Front-end development of an iOS and Android application in Dart and Flutter, collaborating through Git version control.

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, HTML/CSS
Technologies: React, Node.js, Git and GitHub, Flutter/Dart, Code.org
Robotics: FIRST Robotics Competition (FRC) and FIRST Tech Challenge (FTC) program leadership, coaching, and mentoring
Methods: Instructional design, curriculum and assessment design, data analysis, AI applications in education